

The sharp constant in Erdős Problem 457

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Abstract

Let $q(n, k)$ be the least prime not dividing $(n + 1)(n + 2) \cdots (n + k)$. We prove that $q(n, \log n) \geq (1 - \varepsilon) \log n \log \log n / \log \log \log n$ for infinitely many n , improving the constant $\frac{1}{2}$ of Tao to the value 1 predicted by the probabilistic model. The proof is elementary and is formally verified in Lean 4.

1 The result

Erdős and Pomerance [3] asked how large $q(n, \log n)$ (meaning $q(n, \lfloor \log n \rfloor)$) can be; see Problem 457 of [2]. In the discussion of that problem, constructions found by GPT-5.2 Pro (prompted by Barreto) showed $q(n, \log n) \geq C \log n$ infinitely often for every C , and Tao showed

$$q(n, \log n) > \left(\frac{1}{2} - o(1)\right) g(n), \quad g(n) := \frac{\log n \log \log n}{\log \log \log n},$$

for infinitely many n , remarking that $\frac{1}{2}$ is the natural limit of his argument and that $g(n)$ is the expected order.

Theorem 1. *For every $\varepsilon > 0$ there are infinitely many n with $q(n, \log n) \geq (1 - \varepsilon) g(n)$. Equivalently, $\limsup_{n \rightarrow \infty} q(n, \log n) / g(n) \geq 1$.*

The constant is best possible in the probabilistic model: for $k = \lfloor \log X \rfloor$ and $Q = \lambda g(X)$, the expected number of $n \in (X, 2X]$ with $q(n, k) > Q$ is $X \prod_{k < p \leq Q} k/p = X^{1 - \lambda + o(1)}$. An upper bound of the form $q(n, \log n) < (1 - c)(\log n)^2$ is Problem 1181 of [2] and is open.

2 Three lemmas

Write $\theta(x) = \sum_{p \leq x} \log p$ and $\psi(x) = \log \text{lcm}(1, \dots, \lfloor x \rfloor)$. Besides Chebyshev's bounds $\theta(x) \leq x \log 4$ and $\psi(x) \leq C_1 x$ with $C_1 = \log 4 + 4$, we use only the following.

Lemma 1 (Chebyshev). *For every $\delta > 0$, $\theta(x) \leq (1 + \delta)x$ for arbitrarily large x .*

Proof. Legendre's formula gives $\sum_{p \leq N} \lfloor N/p \rfloor \log p \leq \log N! \leq N \log N$, hence $S(N) := \sum_{p \leq N} \frac{\log p}{p} \leq \log N + \frac{\theta(N)}{N} \leq \log N + \log 4$. Partial summation gives $S(N) \geq \sum_{n < N} \theta(n) / (n(n + 1))$. If $\theta(n) > (1 + \delta)n$ for all $n \geq n_0$, the right side is $\geq (1 + \delta) \log N - O(1)$, a contradiction for large N . $\square$

Lemma 2 (Anchored pigeonhole). *Let $\Phi : \{0, \dots, J\} \rightarrow \Omega$ with Ω finite, and let F be a finite set of positive integers with $|\Omega|(|F| + 1) < J + 1$. Then $\Phi(i) = \Phi(j)$ for some $0 \leq i < j \leq J$ with $j - i \notin F$.*

Proof. A fibre B of Φ has more than $|F| + 1$ elements; with $i = \min B$, the $|B| - 1 > |F|$ differences $j - i$ ($j \in B \setminus \{i\}$) cannot all lie in F . $\square$

Lemma 3 (Fold). *Let $2h < p$ and $tL \equiv e \pmod{p}$ with $|e| < h$. Then p divides one of $n + 2, \dots, n + 2h$, where $n = tL - h - 1$.*

Proof. $n + (1 + h - e) = tL - e \equiv 0 \pmod{p}$, and $2 \leq 1 + h - e \leq 2h$. $\square$

3 Proof of Theorem 1

Fix $\varepsilon \in (0, \frac{1}{2}]$ and put $c = (3 + C_1)/\varepsilon$. For an integer M let

$$A = \frac{c \log M}{\log \log M}, \quad K = \lfloor AM \rfloor, \quad h = \left\lfloor \frac{M}{20 \log A} \right\rfloor, \quad L = \text{lcm}(1, \dots, M), \quad \mathcal{P} = \{p : M < p \leq K\}.$$

The code. Let $\kappa(j) = (\lfloor [jL]_p/h \rfloor)_{p \in \mathcal{P}}$, where $[x]_p$ is the least nonnegative residue. It takes at most $C_M = \prod_{p \in \mathcal{P}} (\lfloor p/h \rfloor + 1)$ values. Since $\log(p/h)/\log p$ increases in p ,

$$\log C_M \leq \sum_{p \in \mathcal{P}} \left(\log \frac{p}{h} + \frac{h}{p} \right) \leq \theta(K) \frac{\log(K/h)}{\log K} + \pi(K) \frac{h}{M}.$$

As $M \rightarrow \infty$ we have $K/h \asymp A \log A$, $\log K \sim \log M$, $A \log A / \log M \rightarrow c$ and $\pi(K)h/M = o(M)$. Hence, whenever $\theta(K) \leq (1 + \frac{1}{8c})K$ and M is large, $\log C_M \leq (c + \frac{1}{2})M$.

Good scales. There are infinitely many such M . Indeed $K(M+1) \leq (1 + 7/M)K(M)$ for large M ; given a large x with $\theta(x) \leq (1 + \frac{1}{16c})x$ (Lemma 1), take M maximal with $K(M) \leq x$; then $\theta(K(M)) \leq \theta(x) \leq (1 + \frac{1}{16c})K(M+1) \leq (1 + \frac{1}{8c})K(M)$.

The construction. Let M be a large good scale, $J = \lfloor e^{(c+2)M} \rfloor$ and $F = \{1, \dots, \lfloor e^M \rfloor\}$. Then $C_M(|F| + 1) \leq 2e^{(c+3/2)M} < J + 1$, and Lemma 2 gives $t = j - i$ with $e^M < t \leq e^{(c+2)M}$ and $\kappa(i) = \kappa(j)$; thus $tL \equiv e_p \pmod{p}$ with $|e_p| < h$ for every $p \in \mathcal{P}$. Put $n = tL - h - 1$. Every prime $p \leq M$ divides $L \mid n + h + 1$, and every $p \in \mathcal{P}$ divides one of $n + 2, \dots, n + 2h$ by Lemma 3 (note $2h < M < p$). Moreover

$$M < \log t \leq \log n \leq \log t + \psi(M) \leq (c + 2 + C_1)M =: sM.$$

Since $2h < M \leq \log n$, every prime $p \leq K$ divides $\prod_{i \leq \log n} (n + i)$, so $q(n, \log n) > K \geq AM - 1$.

Conclusion. With $X = \log n \in [M, sM]$,

$$(1 - \varepsilon)g(n) \leq (1 - \varepsilon) \frac{sM(\log M + \log s)}{\log \log M} \leq \frac{cM \log M}{\log \log M} - 1 = AM - 1 < q(n, \log n)$$

for large M , because $c - (1 - \varepsilon)s = 1 + \varepsilon(2 + C_1) \geq 1$. As $\log n \geq M \rightarrow \infty$, this gives infinitely many n . $\square$

Remark. Two points remove Tao's factor $\frac{1}{2}$. The covering block has length $2h \asymp \log n / \log \log \log n = o(\log n)$, yet the cost $\log(p/h)$ of a prime differs from $\log(p/\log n)$ only by $O(\log \log \log \log n) = o(\log \log \log n)$; and the anchored choice $t > e^M$ makes n large at a price e^M that is negligible as $c \rightarrow \infty$. The same code was used in [1] for Problem 684. No prime number theorem is needed: the entropy bound is only required at the scales actually used, which Lemma 1 supplies.

4 Formal verification

Theorem 1, its limsup form, and the Formal Conjectures variant `erdos_457.variants.qnk` are verified in Lean 4 (v4.35.0-rc3) with Mathlib [5] (commit b84a70d) at <https://github.com/jbaelaw/erdos457-lean>. The definition of q is copied verbatim from [4]; the main statement is

```
theorem erdos_457_sharp (ε : ℝ) (hε : 0 < ε) :
  {n : ℕ | (1 - ε) * Real.log n * Real.log (Real.log n) /
    Real.log (Real.log (Real.log n)) ≤ q n (Real.log n)}.Infinite
```

and it depends only on the axioms `propext`, `Classical.choice` and `Quot.sound`.

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References

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